

CSE 331

AUTOMATA AND COMPUTABILITY

Lecture 9 (RE to NFA)

Spring 2026

Today's learning goals

Sipser Ch 1.2, 1.3

- Convert between regular expressions and automata

RE to NFA

regexp to NFA (to DFA)

Lemma 1.55, page 67

- Idea: basic regular expressions are easy to implement as DFA, for inductive step of definition, use closure under regular operations.
- Example: build NFA for the language described by $(00 \cup 11)^*$

	Regular exp	NFA
Base case		
①	$R = \emptyset$ $L(R) = \{ \}$	
②	$R = \epsilon$ $L(R) = \{ \epsilon \}$	
③	$R = a$ $L(R) = \{ a \}$	

Regular exp

NFA

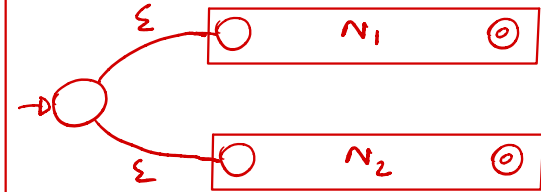
Inductive Case

(4)

$$R = R_1 \cup R_2$$

$$L(R) = L(R_1) \cup L(R_2)$$

$$L(N) = L(N_1) \cup L(N_2)$$

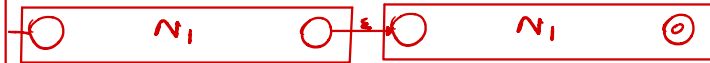


(5)

$$R = R_1 \cdot R_2$$

$$L(R) = L(R_1) \cdot L(R_2)$$

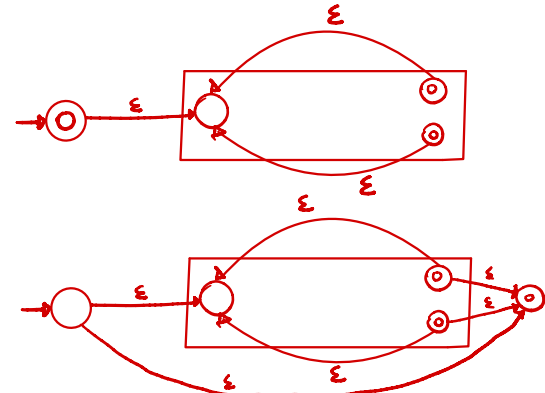
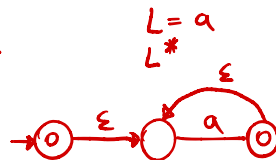
$$L(N) = L(N_1) \cdot L(N_2)$$



(6)

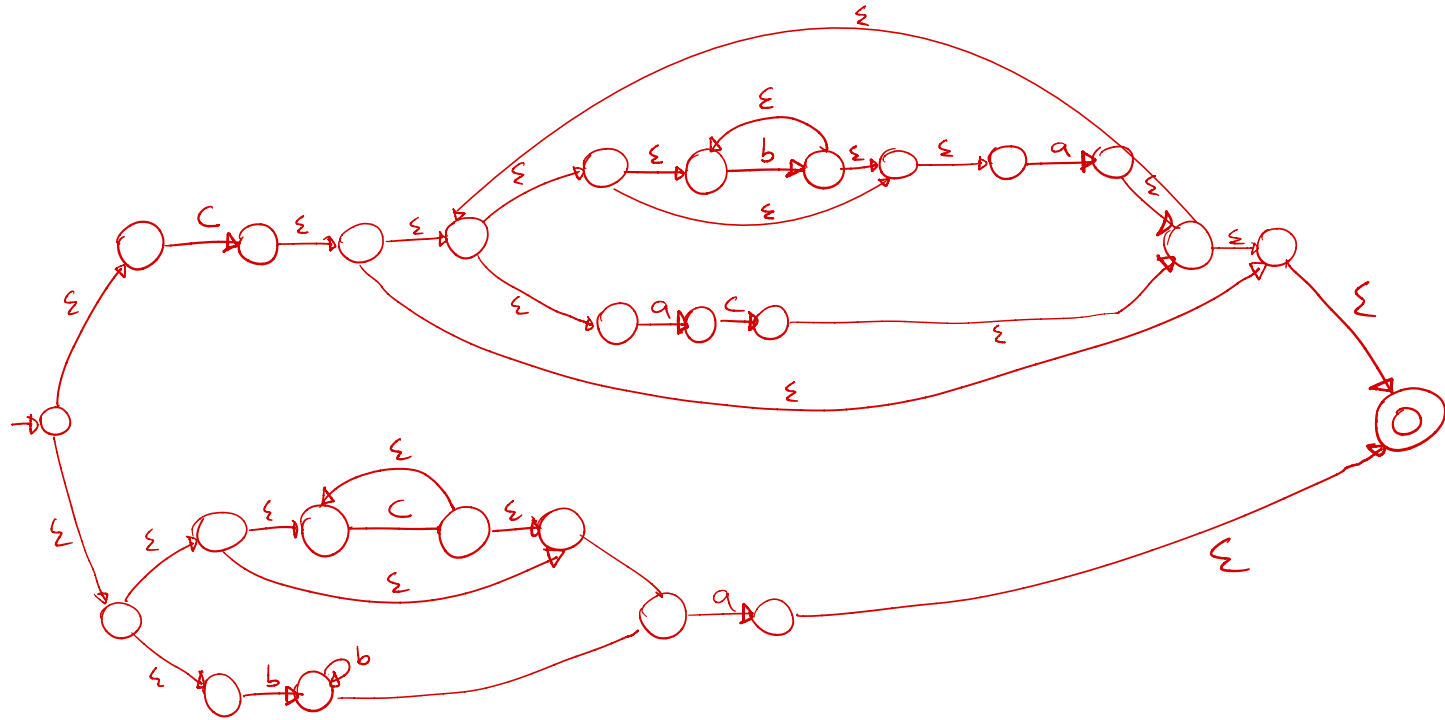
$$R = (R_1)^*$$

$$a^* = \{\epsilon, a, aa, \dots\}$$



$$c(b^*a+ac)^* + (c^*+bb^*)\underline{a}$$

Show direction of arrows.



Thompson Construction rule:

- (i) There can be only one start state.
- (ii) The start state cannot have incoming transition.
- (iii) There can be only one final state.
- (iv) The final state cannot have outgoing edges.
- (v) The number of transition leaving any state is at most 2.